LESSON 7.1
CIRCLE GRAPHS

LESSON INTRODUCTION

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

LEARNING TARGETS

- I can show how an angle measure is related to a portion of a circle.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.9.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What is a central angle?
- How is the central angle measure of a sector related to the fractional part of the whole circle that the sector represents?

LESSON NARRATIVE

Students explore dividing circles into sectors, labeling the fraction and percentage represented by each section. They explore dividing circles into congruent sectors first and then noncongruent sectors. Then, the Guided Instruction connects this exploration to the meaning of central angles. Students identify central angle measures of the sectors and connect the angle measure ratios ($\frac{\text{angle measure}}{360°}$) to the fractional part of the area of the whole circle that the sector represents.

COMPONENT SUMMARY

7.1.1 WARM-UP *(~5 MIN.)*

Compare data represented in circle graph and bar chart

7.1.2 EXPLORATION *(10 MIN.)*

Explore dividing circles into congruent sectors and determine the percentage of the circle represented by each sector

7.1.3 COLLABORATION *(10 MIN.)*

Divide circles into noncongruent sectors and label

7.1.4 GUIDED INSTRUCTION *(15-20 MIN.)*

Measure central angles based on number of congruent sectors

7.1.5 WRAP-UP *(~5 MIN.)*

Students reflect on what they are unsure about and on a question that they have about the lesson

RESOURCES

GLOSSARY

Key Terms:

- Sector
- Angle
- Central angle

Additional Terms:

- Circle graphs
- Vertex

MATERIALS

- Protractors for students
- Straightedges for students
- (Optional) Highlighters
- (Optional) Circle fraction tiles

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to multiply by 100 after dividing the numerator by the denominator when converting a fraction to a percentage.
- Students may struggle to divide circles into thirds and sixths from prior learning on how to use central angles.

THINGS TO CONSIDER

- The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions.
- Consider allowing students to sketch the sections, particularly for thirds and sixths, if they are not sure how to divide the circles exactly. They can revisit their work from the 7.1.2 Exploration and 7.1.3 Collaboration after learning about central angles in the 7.1.4 Guided Instruction. It is not recommended to reverse the order, as the 7.1.2 Exploration will build a stronger understanding than jumping straight into the 7.1.4 Guided Instruction.
- Consider creating and displaying an anchor chart with the instructions, visual models, and examples of determining central angles from sectors.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 7.1.4 Guided Instruction, engage students in comparing and connecting the angle ratios and the fractional parts of the area of the circles. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 7.1.3 Collaboration and 7.1.4 Guided Instruction, students represent sectors as fractions, percentages, and angles. Highlight connections between the concepts and representation to help students build conceptual understanding and procedural fluency.

DIFFERENTIATE INSTRUCTION

The circle graphs in 7.1.2 Exploration provide visual entry points for students dividing and labeling circles into congruent fractional parts. The discussions during 7.1.3 Collaboration provide auditory entry points, as students discuss how to divide and label circles into noncongruent fractional parts, now introduced as sectors.

Additional differentiation opportunities are denoted throughout the lesson guide.

7.1.1 WARM-UP: NFL DRAFT

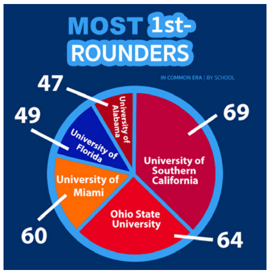
TEACHER MOVES

Notice and Wonder

Ask students what they notice and what they wonder about each data display, which is a better representation of the data, and why.

7.1.1 Warm-Up: NFL Draft

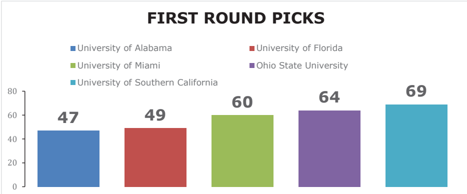
The circle graph shows the number of National Football League's (NFL) first round picks for five schools for the common era ending in 2017.



1. What do you notice about the circle graph?

   Answers will vary. I noticed that the numbers add up to more than 100. I noticed that the sections represent what portion of the whole number of picks are from each school.

2. A data scientist created a bar chart for the same data.



   Describe how the bar chart better represents the data than the circle graph.

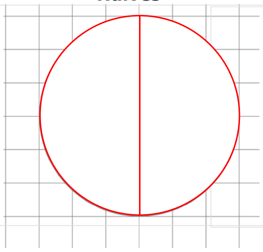
   The bar chart better shows how the number of first round draft picks from each school compare to each other in size.

7.1.2 Exploration: Sectors and Circles

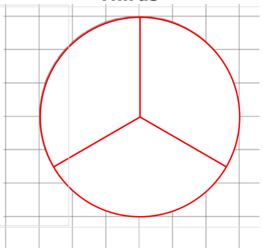
A ***sector*** is a fractional part of the area of a circle.

1. There are six congruent circles drawn below. Use a protractor to divide the circles into sectors that represent halves, thirds, fourths, fifths, sixths, and eighths, as indicated.

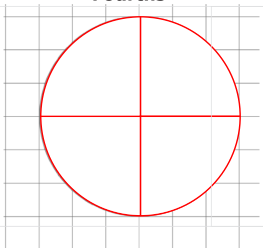
Halves



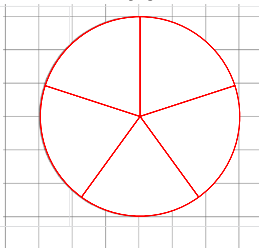
Thirds



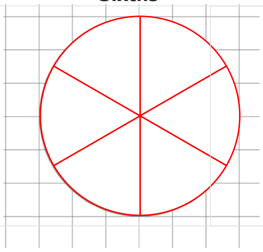
Fourths



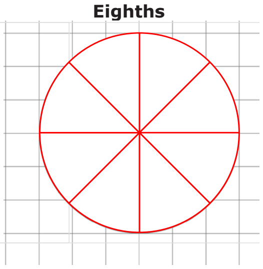
Fifths



Sixths



Eighths



2. Complete the table with the fraction and percentage of the whole circle that each congruent sector represents.

Sector	Fraction	Percentage
Halves	$\frac{1}{2}$	50%
Thirds	$\frac{1}{3}$	≈ 33%
Fourths	$\frac{1}{4}$	25%
Fifths	$\frac{1}{5}$	20%
Sixths	$\frac{1}{6}$	≈ 17%
Eighths	$\frac{1}{8}$	12.5%

7.1.2 EXPLORATION: SECTORS AND CIRCLES

QUESTIONING

Before using a protractor, develop students' reasoning and sense-making skills by asking them to see if students can use a pencil to sketch an approximation of where they estimate the sectors to be. Show students how in addition to using the protractor, they can fold the paper or use the grid lines to check their estimates.

TEACHER MOVES

Observe while students are working to determine if there are particular divisions they have a harder time with than others. Listen for students' discussions on angle relationships and how they think about using the protractor to help, in particular when it comes to thirds and sixths.

TEACHER MOVES

Indicate the place value you want students to round the percentages to as needed. In this case, it would be appropriate to have them round to the nearest whole, as they are beginning to explore this concept. When they begin constructing circle graphs, they may need more precision.

7.1.3 COLLABORATION: DIVIDING CIRCLES INTO SECTORS

QUESTIONING

Ask students to reflect back on the circle graph in the 7.1.1 Warm-Up and see if they label the sectors with the correct fractions and percentages.

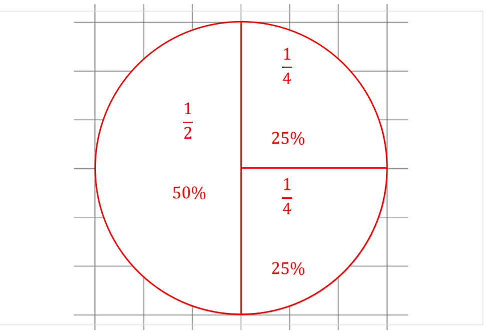
DIFFERENTIATE INSTRUCTION

Students may not recall that the sum of fractions that make up a whole is equal to 1. Encourage students to write an addition expression using the fractions they used to label the circle and find the sum.

7.1.3 Collaboration: Dividing Circles into Sectors

In the Exploration, all of the circles were divided into congruent sectors. Data displayed in **circle graphs** are not typically divided equally among each category.

1. Use the circle on the coordinate grid to answer the following questions.



a. Discuss with your partner how to divide the circle into a half and two quarters.

Answers will vary. I would draw a horizontal line through the circle to make halves and then a vertical line through one of the halves to make two quarters. I would draw a vertical line through the circle and draw a horizontal line through one of the halves to make two quarters.

b. Use a straight edge to divide the circle. Label each sector with the fraction and percentage of the whole circle the sector represents.

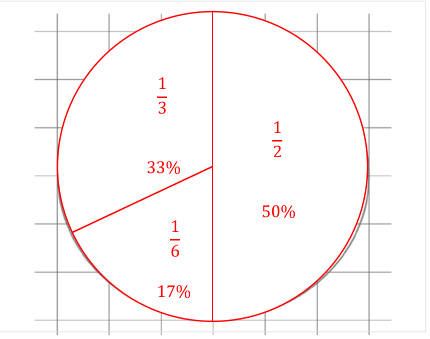
c. What is the sum of all the fractions in the labeled circle?

$\frac{1}{2}+\frac{1}{4}+\frac{1}{4}=1$

d. What is the sum of all the percentages in the labeled circle?

$50\% + 25\% + 25\% = 100\%$

2. Use the circle on the coordinate grid to answer the following questions.



a. Discuss with your partner how to divide the circle into a half, a third, and a sixth.

Answers will vary. I would divide the circle in half and then divide one of the halves into one third and two thirds. I would divide the circle in half and then draw an X across the circle. I would then erase lines, leaving on one third and two thirds in one of the halves.

b. Use a straight edge to divide the circle. Label each sector with the fraction and percentage of the whole circle the sector represents.

c. What is the sum of all the fractions in the labeled circle?

$\frac{1}{3}+\frac{1}{6}+\frac{1}{2}=1$

d. What is the sum of all the percentages in the labeled circle?

$33\% + 17\% + 50\% = 100\%$

3. Consider a circle divided into any number of sectors.

a. What should be the sum of all fractions in order to represent the area of the whole circle?

1

b. What should be the sum of all the percentages in order to represent the area of the whole circle?

100%

7.1.3 COLLABORATION: DIVIDING CIRCLES INTO SECTORS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students may struggle with dividing the circle into thirds and sixths. Consider providing a kinesthetic entry point by allowing students to use circle fraction tile manipulatives to model the divisions. Students can use the tiles to trace the sectors.

Consider also that students can revisit their divisions after the 7.1.4 Guided Instruction by measuring the central angles and making adjustments as needed after developing a stronger understanding.

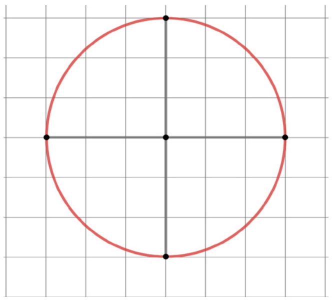
7.1.4 Guided Instruction: Determining Central Angle Measures

An ***angle*** is formed wherever two lines, segments, or rays intersect. Angles are measured in degrees.

A ***central angle*** has its vertex at the center of a circle with radii as its sides.

A diagram of a circle with four congruent sectors is shown.

Use the definition of a central angle to determine how many central angles are in the circle with four congruent sectors.



1. Complete the statements.

Each of the equal sectors form a ○ right ○ straight angle in the center of the circle. A ○ right ○ straight angle is a 90 degree angle. The total sum of the 4 central angles equals ○ 90 degrees. ○ 360 degrees.

7.1.4 GUIDED INSTRUCTION: DETERMINING CENTRAL ANGLE MEASURES

DIFFERENTIATE INSTRUCTION

Students may benefit from a review of the terms, “radii” and “vertex” as the definition for central angle is discussed. Consider providing a visual entry point by highlighting the associated parts of the circle while discussing the terms. After highlighting and labeling a radius and the vertex, draw in a second radius and explicitly note that the two radii form the central angle.

7.1.4 GUIDED INSTRUCTION: DETERMINING CENTRAL ANGLE MEASURES (CONTINUED)

ENRICHMENT

Consider asking students the following:

- Why is the denominator in any angle ratio within a circle equal to 360°?
- How can you check that your angle measures for each sector are correct?
- How can you show that the fraction 1/2 is equivalent to the angle ratio of 180/360?

TEACHER MOVES

Compare and Connect

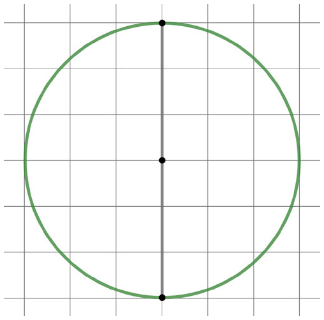
Engage students in comparing and connecting the angle ratios and the fractional parts of the area of the circles from the 7.1.2 Exploration. These discussions will help students gain confidence and understanding.

The measure of each central angle can be written as a ratio of the angle to $360°$. Thus, each sector in the circle divided into fourths takes up $\frac{90}{360}$ of the degrees of the circle.

2. How does this ratio compare to the fraction used to label the sector in the Exploration?

   If you reduce $\frac{90}{360}$, it becomes $\frac{1}{4}$ which is the fraction used when a circle is divided into fourths.

3. A circle that is divided into halves is shown.



   a. What is the measure of one of the central angles?

   $180°$

   b. What is the name of the type of angle that represents the central angle?

   Straight angle

   c. Write the ratio of the measure of the central angle to $360°$.

   $\frac{180}{360}$

   d. How does this ratio compare to the fraction used to label the sector in the Exploration?
   If $\frac{180}{360}$ is simplified, it becomes $\frac{1}{2}$ which is the fraction created when dividing a circle into halves.

4. Answer the following with your partner.

   a. If a circle is divided into any number of congruent sectors, how can the central angle measures be calculated? Summarize your discussion.
   If a circle is divided into a number of congruent sectors, the central angle measures can be calculated by dividing $360°$ by the number of sectors.

   b. Based on your discussion, fill in the table.

Circle Division	Central Angle Measure	Ratio of Angle Measure to $360°$
Thirds	$120°$	$\frac{120}{360} = \frac{1}{3}$
Fifths	$72°$	$\frac{72}{360} = \frac{1}{5}$
Sixths	$60°$	$\frac{60}{360} = \frac{1}{6}$
Eighths	$45°$	$\frac{45}{360} = \frac{1}{8}$

   c. Compare the ratios in the table to the fractions used to label the circles in the Exploration. Discuss what you notice with your partner.
   The ratios, when reduced will equal the fractions from the exploration. (Shown above)

   d. Write a conjecture of how the angle measure ratio relates to the amount of a circle.
   When the angle of a sector is written as a ratio to $360°$, it can be simplified to determine the fractional part of the circle the sector represents.

7.1.5 Wrap-Up: Reflection

Complete each statement based on your understanding of today's lesson.
Answers will vary.

1. I am still unsure about ____________________________
____
____.

2. One question I have that will help me better understand is ____
____
____?

7.1.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

If time permits, consider having students walk around the room to find a partner who can answer their question. Alternatively, you can assign 5-7 student captains to facilitate small groups; students can opt into a group based on what each student captain identifies as their strength.